SDAIA Academy
أكاديمية سدايا

Modular Stackable Learning Paths

مسارات التعلم المعيارية القابلة للتكديس

One architecture — from workforce literacy to expert engineering

AI Engineer & Data Scientist Professional Pathways + Beginner Modules for the Non-Technical Workforce · Catalog v2 · 2026

41	3+1	710	12
Stackable modules	National levels	Learning hours	Credentials

THE BIG PICTURE

One Architecture, Every Audience

A single competency ladder — from everyday AI literacy to national-scale expertise

Badges never expire and carry forward — every step stacks toward the next credential.

L01 · General Workforce

الثقافة والإتقان العام

10 standalone modules · no technical background

Foundation Core

النواة التأسيسية

5 modules · 17 days · open entry

Practitioner · ممارس

Level 1

Contributes under guidance

Specialist · متخصص

Level 2

Delivers production workloads

Expert · خبير

Level 3

Architects & leads · panel capstone

Everyday users ⟶ Technical professionals ⟶ National-scale experts

DESIGN PRINCIPLES

Redesign Approach

From fixed programs to a competency-based, modular, stackable architecture

01 Backward Design

Measurable Bloom-aligned outcomes drive content and assessment in every module.

02 Modular & Stackable

Every module = 2–5 days × 5 hours; standalone value, stacks into certificates.

03 Competency Levels

Practitioner ⟶ Specialist ⟶ Expert, aligned with the SDAIA national classification.

04 Project-Based

40%+ hands-on labs; both tracks conclude with panel-assessed capstones.

05 Industry-Aligned

GenAI engineering, RAG, agentic AI, MLOps/LLMOps, Lakehouse, causal inference.

06 Governed & Scalable

PDPL/NDMO compliance built-in; coding scheme absorbs future tracks.

COMPETENCY LADDER

Three National Levels

Aligned with the SDAIA Academy target-audience classification methodology

LEVEL 1

Practitioner · ممارس

Entry tier. Core technical and functional fluency for professionals and graduates.

EXIT PROFILE

Contributes to AI/data work under guidance

→

LEVEL 2

Specialist · متخصص

Mid tier. Intermediate skills applied to real delivery of models, pipelines, and GenAI apps.

EXIT PROFILE

Delivers production workloads independently

→

LEVEL 3

Expert · خبير

Top tier. Architects, secures, optimises, and operates national-scale systems; leads teams.

EXIT PROFILE

Panel-defended capstone and leadership

Each level closes with a stackable certificate; badges earned never expire and carry forward.

STRUCTURE

Learning Architecture

One shared foundation feeding two role-based professional tracks

Foundation Core · 5 modules · 17 days · النواة التأسيسية

AI & Data Literacy · GenAI & Prompting · Python · Data Wrangling & SQL · Responsible AI & Governance

AI Engineer Track · مهندس الذكاء الاصطناعي

15 modules · 55 days

Practitioner (3)	Applied ML · Deep Learning · SW Engineering for AI
Specialist (6)	NLP · Vision · LLM Apps · RAG · Fine-Tuning · MLOps
Expert (6)	Agentic AI · LLMOps · Security · Optimization · Architecture · Capstone

Data Scientist Track · عالم البيانات

11 modules · 39 days

Practitioner (2)	Statistics · Visualization & Storytelling (+ Applied ML cross-listed)
Specialist (5)	Advanced ML · Time Series · Experimentation & Causal · Data Engineering · Spark
Expert (4)	Decision Science · GenAI Workflows · Streaming · Capstone

Cross-listing: *Applied ML Foundations (SDA-AIE-111) counts toward both tracks — no duplicated teaching.*

AI Engineer Track — Module Portfolio

15 modules from applied ML to agentic AI systems — 55 training days

CODE	MODULE	LEVEL	DAYS	FOCUS
SDA-AIE-111	Applied Machine Learning Foundations	Practitioner	5	ML problem framing and workflow design
SDA-AIE-112	Deep Learning Essentials with PyTorch	Practitioner	4	Neural networks, activations, and backpropagation
SDA-AIE-113	Software Engineering Practices for AI Systems	Practitioner	3	Structuring AI projects and clean architecture
SDA-AIE-211	Natural Language Processing with Transformers	Specialist	4	Text processing, tokenisation, and embeddings
SDA-AIE-212	Computer Vision Systems Development	Specialist	4	Image data, preprocessing, and augmentation
SDA-AIE-213	Large Language Model Application Engineering	Specialist	4	LLM application architecture patterns
SDA-AIE-214	Building Retrieval-Augmented Generation Systems	Specialist	4	RAG architecture vs fine-tuning: when to use each
SDA-AIE-215	Fine-Tuning and Adapting Foundation Models	Specialist	3	Adaptation strategy: prompt vs RAG vs fine-tune
SDA-AIE-216	MLOps: Model Deployment and Automation	Specialist	5	MLOps maturity model and system architecture
SDA-AIE-311	Agentic AI Systems Engineering	Expert	5	Agentic system architectures and components
SDA-AIE-312	LLMOps and Production AI Observability	Expert	4	From MLOps to LLMOps: what changes
SDA-AIE-313	AI Security, Guardrails and Red Teaming	Expert	3	AI threat landscape and OWASP LLM Top-10
SDA-AIE-314	AI Performance, Scalability and Cost Optimization	Expert	3	Performance profiling of inference workloads
SDA-AIE-315	Enterprise AI Solution Architecture	Expert	3	AI solution architecture lifecycle
SDA-AIE-390	AI Engineering Capstone: Production GenAI System	Expert	5	Project scoping and requirements engineering

Data Scientist Track — Module Portfolio

11 modules from statistics to streaming analytics — 39 training days

CODE	MODULE	LEVEL	DAYS	FOCUS
SDA-DSC-111	Statistical Foundations for Data Science	Practitioner	4	Probability and common distributions
SDA-DSC-112	Data Visualization and Storytelling	Practitioner	3	Perception science and visual encoding
SDA-DSC-211	Advanced Machine Learning Methods	Specialist	4	Gradient boosting deep dive
SDA-DSC-212	Time Series Analysis and Forecasting	Specialist	3	Time-series structure and decomposition
SDA-DSC-213	Experimentation, A/B Testing and Causal Inference	Specialist	4	Causal thinking and potential outcomes
SDA-DSC-214	Modern Data Engineering for AI Systems	Specialist	5	Warehouse, lake, Lakehouse: evolution of data architecture
SDA-DSC-215	Big Data Analytics with Spark	Specialist	3	Distributed computing fundamentals
SDA-DSC-311	Decision Science and Optimization Modeling	Expert	3	From predictive to prescriptive analytics
SDA-DSC-312	GenAI-Augmented Data Science Workflows	Expert	3	AI-assisted coding and analysis patterns
SDA-DSC-313	Real-Time and Streaming Data Analytics	Expert	3	Streaming architecture and event-driven design
SDA-DSC-390	Data Science Capstone: From Data to Decision	Expert	5	Problem framing and scoping workshop

ENTRY & ANCHORS

Foundation Core & Flagship Modules

Open entry for all professionals; two existing programmes absorbed as flagships

FOUNDATION CORE — 17 DAYS

CODE	MODULE	DAYS
SDA-FND-101	AI and Data Literacy for Professionals	2
SDA-FND-102	Generative AI and Prompt Engineering Essentials	2
SDA-FND-103	Python Programming for Data and AI	5
SDA-FND-104	Data Wrangling, SQL and Exploratory Analysis	5
SDA-FND-105	Responsible AI and Data Governance	3

Foundation modules are open to all professionals — no prerequisites; completing the core unlocks both professional tracks.

FLAGSHIP MODULES

SDA-AIE-311

Agentic AI Systems Engineering

5 days · Expert · from "Advanced Agentic AI Systems"

ReAct, Plan-and-Execute, Reflection, multi-agent orchestration, MCP integration, guardrails, red teaming — closing with a deployed enterprise capstone.

SDA-DSC-214

Modern Data Engineering for AI Systems

5 days · Specialist · from "Modern Data Engineering"

Lakehouse architecture, Delta Lake & ACID, ELT, streaming, data quality and governance — with a hands-on Mini-Lakehouse build.

CREDENTIALS

Stackable Certification Pathways

Badges ⟶ level certificates ⟶ professional certificates ⟶ micro-specializations

Foundations Certificate
17 days

→

Practitioner Certificate
29 days

→

Specialist Certificate
45 days

→

Expert Professional Certificate
59–65 days incl. capstone

MICRO-SPECIALIZATIONS — STACK ACROSS LEVELS

GenAI Engineering
LLM Apps + RAG + Fine-Tuning + Agentic AI

MLOps & AI Reliability
MLOps + LLMOps Observability + Optimization

AI Security
Governance + Security & Red Teaming + Agentic AI

Analytics & Experimentation
Time Series + Causal Inference + Decision Science

Data Engineering for AI
Lakehouse + Spark + Streaming Analytics

Recognition of prior learning: *placement assessments allow direct entry at Specialist or Expert level.*

BEFORE → AFTER

What Changed from the Current Catalog

Gap analysis outcomes built into the redesign

Standardised durations

All modules now 2–5 days × 5 hours — schedulable, stackable, budgetable. Multi-week programmes decomposed.

Gaps closed

Added prompt engineering, LLM app engineering, fine-tuning/PEFT, AI security & red teaming, LLMOps, streaming, decision science, solution architecture.

Redundancy removed

Single shared Foundation Core; Applied ML cross-listed to both tracks instead of taught twice.

Sequencing fixed

Explicit prerequisites and unlock chains; governance and security precede production deployment.

Programmes absorbed

Agentic AI Systems ⟶ SDA-AIE-311 (Expert flagship); Modern Data Engineering ⟶ SDA-DSC-214 (Specialist flagship) — objectives rewritten as measurable Bloom outcomes.

Assessment upgraded

Every module ends in an applied artefact; both tracks close with panel-defended capstones.

EXTENDING THE ARCHITECTURE DOWN THE LADDER

Beginner-Level Modules for the Non-Technical Workforce

المستوى المبتدئ – الثقافة والإتقان العام (L01)

Awareness · literacy · productivity · safe and responsible everyday use of AI and data — no programming, statistics, or IT background required.

10	2	130	26
Standalone modules	Tracks: AI + Data	Learning hours	Training days

Design Rationale

منهجية التصميم — four commitments behind the beginner tier

Audience-First

01

Zero technical background assumed — and it stays that way. Hands-on work uses everyday workplace tools; success is measured in workplace behavior, not technical skill.

Standalone Containers, Not a Ladder

02

No module is a prerequisite for another. Employees take any module by job need, or several in any order, with light recommended pairings only.

Container Flexibility

03

Outcomes are written at the competency level; named tools live at topic level — so tools, examples, and duration (±1 day) can be refreshed without re-approval.

Catalog Fit — Tier 4

04

The ten containers provide structure and depth to general mastery, integrating today's five short standalone courses into stable and enduring competency domains.

AI Track — Five Modules

الذكاء الاصطناعي لعموم القوى العاملة

CODE	MODULE	العنوان	DAYS	HRS
FAE	AI Essentials for the Workplace	أساسيات الذكاء الاصطناعي لبيئة العمل	2	10
FGP	Generative AI for Workplace Productivity	الذكاء الاصطناعي التوليدي لرفع الإنتاجية	3	15
FPE	Practical Prompt Engineering for Professionals	هندسة الأوامر العملية للمهنيين	2	10
FNA	No-Code AI: Automating Everyday Work	الأتمتة الذكية بدون برمجة	3	15
FRA	Safe & Responsible AI Use in the Workplace	الاستخدام الآمن والمسؤول للذكاء الاصطناعي	2	10

Track total: 12 days · 60 hours

Data Track — Five Modules

البيانات لعموم القوى العاملة

CODE	MODULE	العنوان	DAYS	HRS
FDL	**Data Literacy Essentials**	أساسيات الثقافة البيانية	2	10
FDS	**Everyday Data Skills with Spreadsheets**	مهارات البيانات اليومية بالجداول الإلكترونية	4	20
FDD	**Data-Driven Decision Making**	اتخاذ القرار المبني على البيانات	3	15
FDV	**Dashboards & Data Storytelling**	لوحات المعلومات وسرد القصص البيانية	3	15
FDP	**Data Protection & Privacy (PDPL)**	حماية البيانات والخصوصية في بيئة العمل	2	10

Track total: 14 days · 70 hours

FULL PORTFOLIO AT A GLANCE

Portfolio at a Glance & Next Steps

41	142	710	7	5
Modules	Training days	Learning hours	Certificates	Specializations

RECOMMENDED NEXT STEPS

1. Investigate converting beginner-level modules for non-technicals into self-paced online learning.
2. Add future tracks (AI Product Manager, AI Solution Architect) on the same architecture.